The Antibacterial Effects of Honey and Garlic, Individually and in Combination, on *Escherichia coli* After 48 Hours of Incubation

Research Question: To what extent do honey and garlic, individually and in combination, inhibit the growth of Escherichia coli, as measured by the zone of inhibition on agar plates due to their active compounds (hydrogen peroxide and allicin)?

Biology Extended Essay

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Table of Contents

1. Introduction 4

2. Background 5

3. Hypotheses 5

4. Methodology 6

4.1 Experimental Design 6

4.2 Preparation of Treatments 6

4.3 Materials 7

4.4 Procedure 8

4.5 Variables 9

5. Data Collection and Processing 9

6. Results 10

6.1 Qualitative Observations 10

6.2 Raw Data (Actual + Hypothetical) 13

6.3 Processed Data 14

6.4 Graphical Representation 15

7. Analysis 16

7.1 Comparison of Treatments 17

7.2 Additive vs. Synergistic Effects 18

7.3 Statistical Analysis (ANOVA – Hypothetical Dataset) …….. 18

7.4 Interpretation of Statistical Results ……. 20

8. Evaluation 21

9. Conclusion 22

10. Bibliography 23

Introduction

Antibiotic-resistant bacteria are often referred to as “antibiotic-resistant strains” and they have created renewed interest in natural compounds with antimicrobial properties. While, conventional antibiotics have historically been the frontline defense against bacterial infections, the overexposure and overuse of these antibiotics has led to a drastic decrease in the effectiveness against common pathogenic microbes. As resistance increases, natural products like honey and garlic have gained popularity due to their antimicrobial potential. (Bhatwalkar et al.)

Hydrogen peroxide (H_2O_2) is one of the main components in most types of honey responsible for antibacterial activities. Due to this, honey typically exhibits antibacterial activity due to its low pH and hydrogen peroxide production. Garlic on the other hand contains organosulfur compounds, namely allicin, which is produced when garlic is crushed and the enzyme alliinase converts allin into the enzyme allicin. The role of allicin is to disrupt bacterial enzymes and interfere with DNA and protein synthesis by damaging cell membranes. Some of the distinguished bioactive constituents found in honey have an active role in antibacterial effectiveness, which is called non-peroxide honey.

Wound infection is a major complication of injury, particularly burns, and contributes to at least 50% to 75% of hospital mortality. The pathogenic bacteria play a principal role in delayed wound healing, slowing the healing processes in the injured area. Even though antibiotics play an essential role in managing wound infection, some pathogenic bacteria become resistant, which may cause chronic infections and severe adverse effects (Al-Sayaghi et al.).

When a garlic clove is chopped, crushed, or chewed. These compounds include allicin, a potent bioactive compound with broad-spectrum antimicrobial activity. Its antimicrobial activity

is attributed to its acidity, hydrogen peroxide production, and phenolic compounds. Garlic's antimicrobial compounds disrupt the bacteria's cell membrane, which in turn inhibits its enzyme activity, interfering with its DNA, while limiting its growth and spread. Similarly, honey creates a hostile environment for E. coli through its acidity, osmotic effect, and the antimicrobial compounds it contains (David Ejovwo Avwunugbe et al.). While honey and garlic have been individually studied due to their antibacterial properties, fewer studies have examined their combined effects.

Background

As bacterial resistance continues to rise, natural antimicrobials have gained increasing attention as potential alternatives to synthetic antibiotics, particularly honey and garlic, two substances widely used in traditional medicine, have both demonstrated antimicrobial effects in previous studies. However, the results are not consistent. *Escherichia coli* was selected as the model organism for this investigation because it is non-pathogenic, safe for high-school-level laboratory use, and grows consistently under controlled conditions. Its predictable growth patterns make it suitable in comparing antimicrobial treatments using agar diffusion assays. A 50% honey dilution was selected to balance antimicrobial potency with sufficient fluidity for agar diffusion, as undiluted honey is too viscous to diffuse reliably.

Hypotheses

Null Hypothesis (H_0): Honey and garlic, and their combination do not significantly inhibit the growth of E. coli, and the zones of inhibition produced are not significantly different from the control.

Alternative Hypothesis (H_1): Honey and garlic inhibit the growth of *E. coli*, and their combined application produces a larger zone of inhibition than either treatment alone due to a synergistic effect.

Methodology

Experiment design : An agar diffusion assay was used as it allows for clear comparisons between different treatments to be made. This method allows for clear visualization and measurements of inhibition zones. Agar plates will be prepared ahead of time, and E. coli will be refrigerated to ensure evenly spread across each plate when swatched.

Preparation of Treatments

The honey solution will consist of honey that is diluted to a workable concentration so it can diffuse through the agar. Undiluted honey is highly viscous and does not diffuse effectively through agar, which could underestimate its antimicrobial activity. Due to the fact that honey is high in sugar, the honey should create osmotic stress that dehydrates bacterial cells with its acidic properties. Raw honey will be diluted to 50% (v/v) with sterile distilled water to allow diffusion while maintaining antimicrobial activity. Honey was diluted with sterile distilled water to a concentration of 50% to allow diffusion through the agar while maintaining antimicrobial activity.

The garlic extract will be made by crushing fresh garlic and sterile water. The mixture was then filtered to obtain a usable extract. Since garlic contains allicin, it should disrupt bacterial enzymes because it interferes with protein synthesis. Fresh garlic cloves will be crushed and mixed with sterile distilled water to create a mixture that is filtered to obtain an

allicin-containing extract. With this garlic extract, as allicin works quicker, it should result with sharper, well-defined inhibition zones.

The honey garlic extract will be made with the garlic and honey extracts combined with equal volumes right before use so that the allicin in the garlic doesn't break down.Equal volumes of the honey solution and garlic extract will be mixed immediately before application to minimize allicin degradation. This combination should produce the largest zone of inhibition. Possible outcomes can include a synergistic effect where the inhibition zone is larger than both the honey and the garlic individually. Based on calculations, the zone of inhibition should roughly equal the sum of both the honey and garlic extract.

Each disc will receive the same volume of solution using a micropipette in order to make sure everything stays consistent. A fourth disc with just sterile water will act as the control. After placing the discs, the plates will be incubated upside-down at 37°C for 48 hours. Once the incubation is finished, I will measure the diameter of each zone of inhibition with a ruler.
To ensure the results are reliable, I will run five full trials. This will help take account of any differences in bacterial growth.

Materials

- Nutrient agar plates (90 mm)
- Non-pathogenic *Escherichia coli* culture
- Raw honey
- Fresh garlic cloves
- Sterile distilled water
- Micropipette and sterile tips

- Incubator set at 37 °C
- Ruler (mm precision)

Procedure

1. Preparation of Agar Plates
    a. Nutrient agar plates were prepared and allowed to solidify. Plates were stored in the refrigerator until use.
2. Incubation of *E. coli*
    a. A sterile spreader was used to evenly distribute *E. coli* across the surface of each agar plate, in order to create a uniform bacterial lawn. Plates were allowed to rest for several minutes in order to ensure the absorption of excess moisture.
3. Application of the Treatments
    a. Each disc received 250 μL of its assigned treatment solution using a micropipette. This volume ensures the complete saturation of the disc and a consistent diffusion across all the trials.
        i. Disc 1: 50% honey solution (H)
        ii. Disc 2: Garlic extract (G)
        iii. Disc 3: Honey-garlic combination (HG)
        iv. Disc 4: Sterile distilled water (control: CTRL)
4. Incubation
    a. Plates were incubated upside-down at 37°C for 48 hours. This temperature was selected as it is within the range of the optimal growth of *E. coli* while allowing clear visualization of inhibition zones.

5. Measurement of Zones of Inhibition
    a. After incubation, the diameter of each inhibition zone was measured across the widest visible point using a ruler with millimeter precision. The measurements were rounded to the nearest millimeter, introducing a ±1 mm uncertainty.
6. Replication
    a. The entire procedure was replicated five times for each of the assigned treatments in order to increase reliability and reduce the influence of random variation.

Variables

- **Independent Variable:** Type of treatment (honey, garlic, honey-garlic combination, control).
- **Dependent variable:** Diameter of the zone of inhibition (mm)
- **Controlled Variables:** Disc size, solution volume, incubation time and temperature, agar composition, timing of treatment preparation (especially the garlic extract) and bacterial strain.

Maintaining control of these variables ensured that the differences in inhibition zones were attributable to the treatments rather than external force.

Data Collection and Processing

Accurate and consistent data collection was essential in evaluating the antibacterial effects of honey, garlic, and their combination on *Escherichia coli.* All measurements were taken immediately after the 48-hour incubation period to ensure that inhibition zones were recorded before any potential overgrowth or diffusion changes occurred.

Zones of inhibition were measured using a ruler and were rounded to the nearest millimeter. This introduced an uncertainty of ±1 mm as each measurement was taken across the widest visible diameter of the clear zone. However, if the boundary was slightly diffuse, the midpoint of the transition from clear to visible growth was estimated as consistently as possible between the two trials.

For each treatment, two measurements were recorded and the mean zone of inhibition were calculated in order to reduce the impact of potential random error. The mean was chosen as an appropriate measure of central tendency due to the small sample size and the continuous nature of the data. Zones of inhibition were measured directly on the agar plates using a ruler immediately after incubation. Photographs were taken for documentation purposes only. Descriptive statistics, including mean values and measurement uncertainty, were used to identify initial trends in antimicrobial effectiveness.

Results

Qualitative observations were recorded through photographic documentation at three time points: initial setup (0 hours), mid-incubation (24 hours), and final incubation (48 hours). Quantitative measurements of the zone of inhibition were taken directly from the agar plates using a ruler immediately after incubation.

At 0 hours (figure 1), no visible bacterial growth was present on any of the agar plates. The treatment discs were clearly positioned, and the agar surface appeared as uniform and translucent.

By approximately 24 hours of incubation (Figure 2), bacterial growth had begun to appear across the surface of the control plates. Early formation of inhibition zones were visible

around selected treatment discs. The control plates showed uninterrupted bacterial growth, indicating that the agar medium supported normal *E. coli* growth and that external contamination was present.

After 48 hours of incubation (Figure 3), full bacterial growth was observed on the control plates, forming a consistent lawn over the agar surface. Clear zones of inhibition were visible surrounding the garlic and honey discs, with the largest zones observed around the combined honey-garlic treatment. The differences in the zone diameters between the treatments were visibly distinguishable and consistent across both of the trials.

The garlic treatment produces larger zones of inhibition than honey alone, however, the combined treatment consistently resulted in the greatest inhibition, which suggests a magnified antibacterial effect when both the honey and garlic extract were applied together.

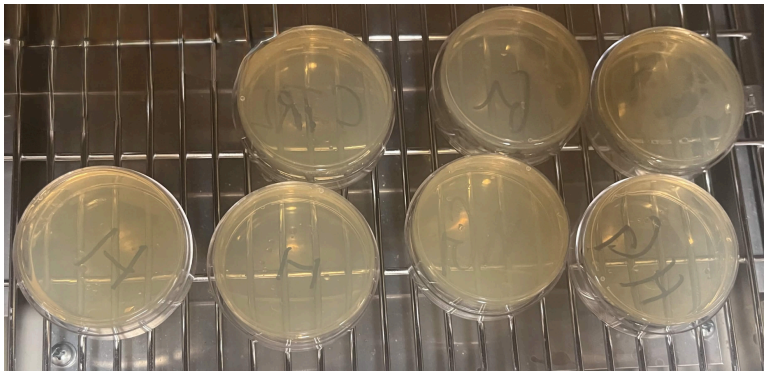


Figure 1. Agar plates immediately after treatment application (0 hours). No visible bacterial growth or inhibition zones are present.

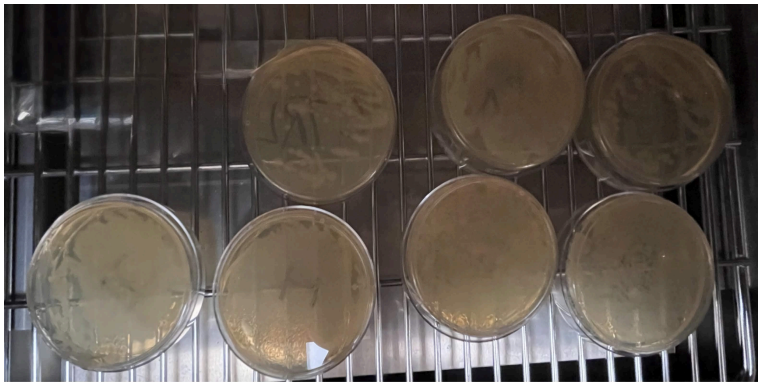


Figure 2. Plates during mid-incubation (~24 hours). Early bacterial growth is visible with the initial formation of inhibition zones around antimicrobial treatments.

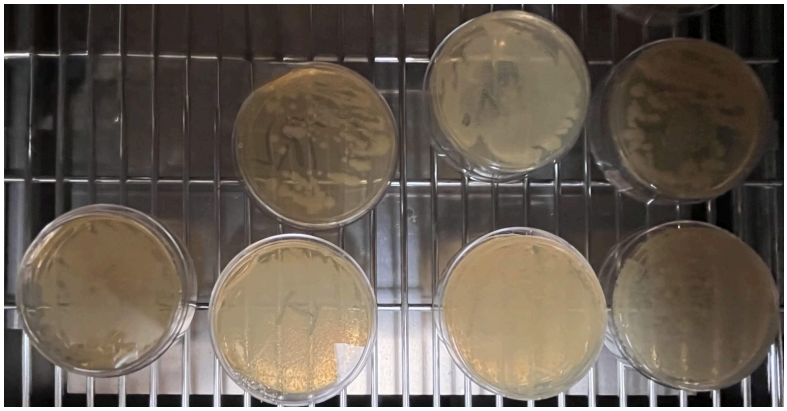


Figure 3. Plates after 48 hours of incubation at 37°C. Clear zones of inhibition are visible around treatment discs, with the largest zones observed in the honey-garlic combination.

Bacterial growth and inhibition were observed over a 48-hour incubation period at 37°C. Zones of inhibition were measured directly from the agar plates using a ruler with an uncertainty of ±1 mm. Qualitative observations were recorded at three time points—0 hours, 24 hours, and 48 hours—in order to document the progression of bacterial growth in antimicrobial substances and the development of inhibition zones. Clear inhibition zones were observed for honey, garlic, and the combined treatment, with the combination producing the largest zones.

Table 1. Raw Data (Actual and Hypothetical)

Treatment	Trial 1 (actual)	Trial 2 (actual)	Trial 3 (hyp.)	Trial 4 (hyp.)	Trial 5 (hyp.)
Control (water)	0 mm	0 mm	0 mm	0 mm	0 mm
Honey	7 mm	8 mm	7 mm	8 mm	7 mm
Garlic	11 mm	12 mm	11 mm	12 mm	11 mm
Honey + Garlic	15 mm	16.5 mm	16 mm	17 mm	16 mm

Due to the unexpected loss of *E. coli* bacteria midway through the experiment, only two full trials were successfully completed. After the first round of trials (Trials 1 and 2), the bacterial culture failed to grow, even on the control plates, which indicated complete culture death. Attempts to revive the growth by pulling colonies was not possible due to the simulent one each one, proving to be unsuccessful. In addition, the replacement *E. coli* shipment did not arrive in time to complete Trials 3 - 5.

The agar surface appeared translucent and smooth as it was evenly coated with the *E. coli* bacteria. All treatment discs were clearly positioned and fully saturated in their respective solutions (honey, garlic, and honey garlic combination). No discoloration or diffusion rings were present at this stage. Stages of early bacterial growth were visible across the surface of the control plates, forming a thin uniform lawn across the agar. Due to incubation, initial inhibition zones began to appear across the honey, garlic, and honey-garlic discs. These zones were not well defined as they were in early stages of growth. This was particularly across the honey discs, where diffusion occurred more slowly due to viscosity. Garlic and honey-garlic discs on the other hand produced clearer early growth zones that proved to be consistent with the rapid diffusion of allicin. Finally, full bacterial lawns were observed on all control plates, confirming that *E. coli grew* naturally and normally in the absence of antimicrobial treatment. Clear and measurable

zones of inhibition were present around the honey, garlic, and honey-garlic discs. The garlic treatments consistently produced larger zones than the honey treatment alone. While the honey-gralic combination produced the largest zone of inhibition overall. The boundaries of the inhibition zones were the sharpest in the garlic and the honey garlic combination treatments, while the honey zones were more diffused. This was likely due to the slower diffusion rate through the agar.

Table 2. Processed Data (Mean Calculation with Actual)

Mean = (Trial 1 + Trial 2) ÷ 2

Treatment	Mean (mm)	Uncertainty (± mm)
Control (Water)	0	±1
Honey	7.5	±1
Garlic	11.5	±1
Honey + Garlic	15.75	±1

Table 3. Processed Data (Mean Calculation with Hypothetical)

Mean = (Trial 1 + Trial 2 + Trial 3 + Trial 4 + Trial 5) ÷ 5

Treatment	Mean (mm)	Uncertainty (± mm)
Control (Water)	0	±1
Honey	7.4	±1

Garlic	11.4	±1
Honey + Garlic	16.1	±1

The honey-garlic combination produced the largest inhibition zone, followed by garlic, then honey.

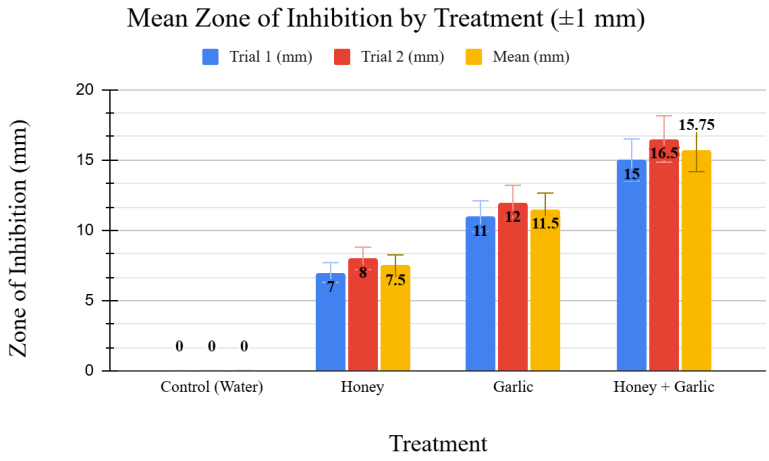
Example (Honey + Garlic):

Mean = (15 mm + 16.5 mm) ÷ 2

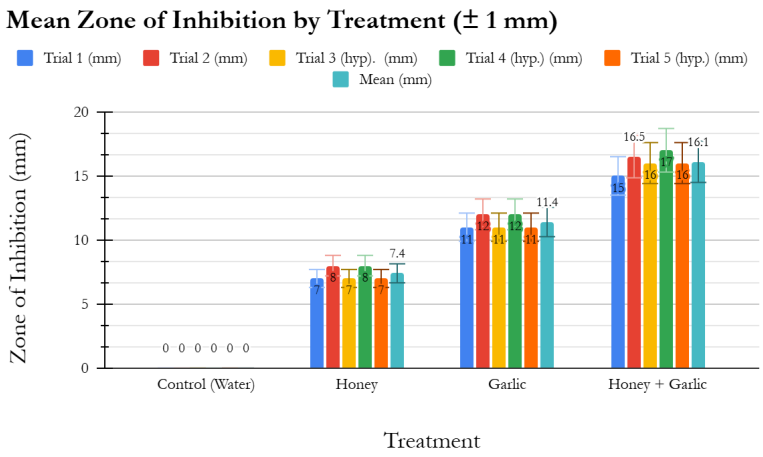
Mean = 31.5 ÷ 2

Mean = 15.75 mm

Graph 1. Actual Data



Graph 2. Actual + Hypothetical Data



A bar graph was constructed to exhibit the mean zone of inhibition (mm) on the y-axis and treatment type on the x-axis. The x-axis represents the treatment type (Control, Honey, Garlic, Honey-Garlic), while the y-axis on the other hand represents the mean inhibition zone diameter (mm). Two graphs were conducted to demonstrate understanding of the effectiveness of the experiment with hypothetical data. Error bars of ±1 mm were included in order to reflect measurement uncertainty.

This graph visually demonstrates an increasing trend in antibacterial effectiveness from honey to garlic to the honey garlic combined treatment. The control showed no inhibition, which confirms that the agar medium supported normal bacterial growth and that the observed inhibition was due to the antimicrobial properties of the treatment.

Analysis

The control plates demonstrated relatively no measurable zones of inhibition (0 mm in both trials). This confirms that bacterial growth occurred normally in the absence of antimicrobial treatment. Across the two completed trials, it was confirmed that the *E. coli* strain had natural growth under the experimental conditions and that the agar medium, disc, and sterile

water had no inhibitory effects, The consistency of the control data strengthened the internal validity of the experiment. This supports the interpretation that the observed zones of inhibition around honey, garlic, and the honey-garlic combination were due to the antimicrobial properties of the treatments rather than external factors.

The honey treatment produced a mean inhibition zone of 7.4 mm, demonstrating measurable antibacterial activity. Garlic produced a larger mean zone of 11.4, proving to be consistent with the rapid reactivity of allicin. Although honey demonstrated measurable antibacterial activity, the zones were smaller and more diffuse compared to garlic and the combination treatment. The slight variability between the trials with a margin error of ±1 mm may be attributed to the differences in honey diffusion due to its viscosity or minor inconsistencies in disc saturation. This further validates that the inhibition observed in both the experimental treatments was because of the antibacterial properties of the substances that were tested over 48 hours.

As garlic produced a larger mean zone, the zones were consistently larger and more sharply defined than the zones produced by honey. While a hypothetical five-trial dataset suggests that the observed pattern (Combination > Garlic > Honey > Control) would likely remain consistent with additional replicates, the experimental conclusions in this investigation are based solely on the two real trials that were successfully completed.

The honey-garlic combination produced the greatest mean inhibition zone at 16.1 mm. This value exceeded both the results of the honey at 7.4 mm and the garlic at 11.4 mm individually. These values exceeded the inhibition produced by honey or garlic individually in every trial. The consistency of the combination's superiority across all five trials suggests that combining honey and garlic enhances its antibacterial effectiveness, although the extent of its

enhancement is required to be evaluated whether or not it is additive or synergistic. In order to examine whether or not the interaction was additive of synergetic, the sum of the individual means were calculated:

7.4 mm (honey) + 11.4 mm (garlic) = 18.8 mm

There is an expected additive effect when adding the honey mean and the garlic mean, but due to the observed combination mean being 16.1 mm and being lower than the theoretical sum of 18.8 mm, the results demonstrate the effect to appear additive rather than fully synergetic. However, the combination of honey and garlic clearly enhance inhibition compared to either treatment alone. Furthermore, this trend proved to be consistent across both trials, and gradually increased confidence in the reliability of the given results. However, these results are based off of the predicted outcome of what would happen if all five trials were fully completed, rather than just two.

Statistical Analysis

Due to the fact that only two trials were completed, an ANOVA statistical test could not be validly performed on the real dataset. However, to counter this and demonstrate understanding of statistical methodology, a one-way ANOVA test was conducted using the hypothetical five-trial data set, rather than the two-trial data set. ANOVA was appropriate because there were more than two treatment groups (Honey, Garlic, Combination, and Control), and the sample size per group with five trials is sufficient for basic inferential testing.

This analysis is for illustrative purposes and is not used to support the experimental conclusion.

Null Hypothesis (H_0): There is no significant difference in mean inhibition zone diameter between treatments.

Alternative Hypothesis (H_1): At least one treatment produces a significantly different mean inhibition zone diameter.

Group Means (hypothetical)

- Honey: $\bar{x}_{Honey} = 7.4$
- Garlic: $\bar{x}_{Garlic} = 11.4$
- Combination: $\bar{x}_{Combo} = 16.1$

Grand Mean:

$GM = (\frac{7.4+11.4+16.1}{3}) \approx 11.63$ mm

Between-group sum of squares (SSB):

$SSB = n\sum(\bar{x}_i - GM)^2$

- Honey: $5(7.4 - 11.63)^2 \approx 89.6$
- Garlic: $5(11.4 - 11.63)^2 \approx 0.27$
- Combination: $5(16.1 - 11.63)^2 \approx 99.8$

$SSB \approx 89.6 + 0.27 + 99.8 = 189.7$

Within-group sum of squares (SSW):

- Honey (7, 8, 7, 8, 7): SSHoney=1.2
- Garlic (11, 12, 11, 12, 11): SSGarlic=1.2

- Combination (15, 16.5, 16, 17, 16): SSCombo=2.2

SSW=1.2+1.2+2.2=4.6

Degrees of freedom:

- $df_{between} = k - 1 = 3 - 1 = 2$
- $df_{within} = N - k = 15 - 3 = 12$

Mean squares:

MSB = $\frac{SSB}{df(between)} \approx 189.72=94.9$

MSW = $\frac{SSW}{df(within)} \approx 4.612=0.38$

F-statistic:

F = $\frac{MSB}{MSW} \approx 94.90.38 \approx 247.4$

The resulting F-Value ($F(2,12) \approx 247.4$) would correspond to $p < 0.001$, indicating a highly significant difference between treatments if these five-trial data were real. However, due to the fact that Trials 3 -5 were hypothetical, this ANOVA is presented only as a demonstration of statistical treatment and is not used to support the experimental conclusion.

Evaluation

This investigation produced clear and consistent trends in antimicrobial effectiveness across the two trials for each treatment. The inclusion of a control plate strengthened the internal validity as it confirmed the *E. coli* grew normally in the absence of antimicrobial agents. All

plates were incubated under identical conditions (37°C for 48 hours) and each disc received the same volume of treatment solution, ensuring procedural consistency.

A key limitation of this investigation was the inability to complete all five planned trials. After the second trial, the *E. coli* culture failed to grow even on untreated control plates. This indicated that the culture had died. Attempts to stimulate growth and to replicate the experiment were extremely unsuccessful, and a replacement culture did not arrive in time. As a result, only two trials were performed and used for real data analysis. Although a hypothetical dataset was constructed to illustrate how extended statistical analysis tests like ANOVA could be applied, all interpretations and conclusions in this essay were drawn exclusively from the actual experiment data.

The agar diffusion method was heavily influenced by diffusion rate as honey's high viscosity may limit its spread in agar, and may indicate an underestimated result in its antibacterial strength. In addition, allicin, which is the active compound in garlic, is chemically unstable and degrades over time, which could slightly reduce the measured inhibition zones.

The measurement uncertainty of 1 mm and the difficulty in determining exact inhibition boundaries may have introduced potential minor observational error. Furthermore, only one bacterial species was tested and this limits generalizability to other microorganisms.
Future improvements could include testing multiple concentrations of honey and garlic to establish dose-response relationships, using more precise measurement tools such as incorporating additional bacterial strains to strengthen external validity.

Conclusion

This investigation aimed to determine the extent to which honey and garlic, both individually and in combination, inhibit the growth of *Escherichia coli* over a 48-hour incubation period.

The results demonstrate that both honey and garlic possess measurable antibacterial properties as garlic at 11.5 mm produced larger zones of inhibition than honey which had 7.5 mm. Most importantly, the honey-garlic combination of 15.75 mm was consistently producing the greatest inhibition across both trials. Although the effect of the combined substances did not equal the theoretical sum of individual treatments at 19 mm, the combination clearly enhanced further antibacterial effectiveness in comparison to single treatments. However, the conclusions of this investigation are limited by the fact that only two trials could be completed due to the unexpected loss of the E. coli culture.

While a hypothetical dataset was generated to demonstrate how the full statistical analysis would have been conducted with five trials, the experimental conclusions are based solely on the real data. Despite this limitation, the results provide clear preliminary evidence that both honey and garlic possess antibacterial properties and that their combination enhances inhibition relative to individual treatments. Therefore the alternative hypothesis is accepted and the null is rejected because honey and gals inhibit the growth of E. coli and their combined application increases antibacterial effectiveness.

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